

Product Proposal: Reel or Real? A Context-Aware Deepfake Detection and Verification Framework

Introduction

Deepfakes, hyper-realistic synthetic videos generated by advanced AI, have emerged as a significant threat to digital integrity and public trust. By swapping faces or mimicking the voices of real people, malicious deepfake videos can spread misinformation, facilitate fraud, and erode the credibility of legitimate media. In response to this threat, Reel or Real? proposes a novel context-aware deepfake detection and verification framework. In our context, “context-aware” means the system goes beyond analysing pixel-level patterns; it integrates multiple modalities and sources of information, e.g. visual cues, audio signals, and metadata about content origin, to make a more informed judgment. The goal is to detect AI-generated videos more robustly and transparently than current methods. By leveraging a combination of innovative deep learning and provenance analysis, this framework aims to serve as a reliable tool to distinguish what is real from what is fabricated, thereby helping restore trust in digital media.

Problem Statement

Despite significant research in deepfake detection, current solutions exhibit important limitations. Many detection models suffer from poor generalisation to unseen data, limited multimodal integration, and low interpretability of decisions. For example, convolutional neural network (CNN) detectors like Xception and ResNet perform well on curated benchmark datasets but often fail when faced with real-world forgeries. Newer transformer-based models improve temporal modelling of video but tend to operate as “black boxes,” making their decisions difficult to interpret and trust.

Furthermore, early deepfake datasets (e.g., FaceForensics++, Deepfake Detection Challenge) lacked diversity and complexity, reducing models’ robustness when encountering varied inputs. More recent datasets (KoDF, HAV-DF, WildDeepfake) offer broader demographic coverage and complex scenarios, and specialised sets like FakeAVCeleb include challenging audio-visual combinations. However, even with better data, critical gaps remain in current approaches:

- **Generalisation:** How to ensure the detector remains accurate on unseen deepfakes and diverse datasets (addressing distribution shifts and new attack methods).
- **Multimodal Deepfakes:** How to jointly analyse audio and visual streams (and other context) to catch forgeries that single-modal models miss.
- **Explainability:** How to provide transparent, interpretable results, so that end-users and decision-makers understand why a video was flagged as fake.
- **Provenance Verification:** How to incorporate source verification (e.g., watermarks or trusted metadata) to double-check authenticity and guard against tampering.
- **Ethical AI & User Trust:** How to design the system to be ethically responsible (avoiding misuse, protecting privacy) and user-friendly, fostering trust and adoption.

These challenges define the scope of our project. They illustrate why a more context-aware, multimodal, and explainable approach is essential for the next generation of deepfake defence tools.

Proposed Solution

To address the above challenges, the project proposes a unified, context-aware deepfake detection framework that combines advanced machine learning with multimodal analysis and user-centric design. The solution integrates multiple components to enhance detection accuracy, generalisation, and transparency:

- **Hybrid Spatial-Temporal Modelling:** The framework uses a hybrid architecture of CNNs and transformers to capture both fine-grained spatial details in video frames and long-range temporal patterns across frames. This dual approach is intended to improve generalisation to diverse video manipulations by learning comprehensive video representations.
- **Audio-Visual Fusion:** In addition to video, the system analyses audio tracks using features like Mel-Frequency Cepstral Coefficients (MFCCs). By aligning audio patterns with corresponding visual frames, the system can detect inconsistencies in synchronisation or content that often indicate deepfake manipulation. Fusing audio and visual cues enables the detection of subtle mismatches that single-modality detectors might miss.
- **Explainable AI Integration:** The framework incorporates explainability techniques directly into its inference pipeline. For each analysed video, visual explanation heatmaps (e.g., Grad-CAM) highlight regions of frames that most influenced the detector's decision, and transformer attention maps illustrate how temporal sequences contributed. A confidence score accompanies each verdict to quantify the model's certainty. These features allow users and auditors to understand why a video was flagged as genuine or fake, improving transparency.
- **Provenance and Metadata Verification:** Beyond analysing content, the system checks the integrity of the video's metadata and provenance. It adheres to emerging standards like the Coalition for Content Provenance and Authenticity (C2PA) and can detect or leverage cryptographic watermarks embedded in media. This ensures that even if a deepfake escapes detection by content analysis, discrepancies or flags in the media's provenance record can raise alerts. Combining data-driven detection with source verification provides a more robust defence against fakery.
- **User-Centric Interface:** All detection results and explanations are presented through a user-friendly interface designed for clarity and accessibility. Non-technical users (e.g., journalists, fact-checkers) can upload a video and receive a clear verdict (real or fake) along with visual cues (heatmaps) and source authenticity information. By prioritising usability, the tool enables decision-makers to act on its outputs confidently.

This multifaceted solution directly targets the identified gaps: it improves generalisation via a richer model and diverse training data, exploits multimodal inconsistencies, provides human-interpretable explanations, verifies content provenance, and delivers results in an understandable format.

System Architecture

The system's architecture follows a modular design with three primary layers: (1) Data Processing, (2) Multimodal Detection Model, (3) Output & Verification. Below, we describe each in more detail:

- **Input Processing:** Videos are ingested and pre-processed (e.g., frame extraction and resizing, audio track isolation). Key frames might be sampled from the video to balance performance and thoroughness.
- **Feature Extraction (Vision):** A CNN backbone (e.g., Xception or ResNet family) processes each video frame to extract spatial features. This module learns delicate details such as facial artefacts, lighting inconsistencies, or blending artefacts that often distinguish deepfakes.
- **Feature Extraction (Audio):** In parallel, the audio track is converted into spectral features (MFCCs or similar). Short audio segments aligned with the frames are analysed to capture voice characteristics, background noise patterns, or anomalies in speech that could suggest tampering.
- **Temporal Sequence Modelling:** The transformer module receives the sequence of frame-level features (and optionally audio features) and models the temporal dynamics. This component can capture movements or lip-sync patterns over time, learning how legitimate videos evolve versus how deepfakes may exhibit telltale irregularities in motion or timing.
- **Multimodal Fusion and Classification:** The outputs from the visual CNN, audio analysis, and transformer are combined in a fusion layer. The system integrates these signals to produce a final classification (real or fake). A classification head (such as a fully connected neural network) uses the fused features to output a probability or label for authenticity.
- **Explainability and Metadata Module:** Alongside classification, the system generates an explanation for the decision. The Grad-CAM technique is applied to the CNN to produce heatmaps over frames, and the transformer's attention weights are visualised to indicate which frames or segments were most influential. Meanwhile, the metadata verification sub-module checks for any cryptographic signatures or C2PA metadata in the video file that confirm authenticity or reveal tampering. Results from this module (e.g., "authenticity metadata valid" or "no watermark found") are factored into the overall output.
- **User Interface Layer:** The final outputs – authenticity decision, confidence score, explanation visuals, and provenance details – are compiled and presented to the user through a graphical interface. This interface allows users to review the evidence behind the classification (for instance, seeing the highlighted facial regions that led to a "fake" verdict) and to verify source information. The UI is designed to be intuitive, so stakeholders can make informed judgments without needing ML expertise.

This architecture is implemented using Python and deep learning frameworks (PyTorch/TensorFlow), ensuring flexibility for future improvements. The modular design also means that components (e.g., the CNN backbone or the transformer model) can be updated as new algorithms emerge, without overhauling the entire system.

Datasets & Ethics

Datasets: We will train and evaluate the system on a curated set of public deepfake datasets chosen to cover a wide range of scenarios and ensure generalisation. The primary datasets are:

- Deepfake Detection Challenge (DFDC): A large-scale public dataset of deepfake and real videos, offering diverse subjects and settings. DFDC provides a broad foundation for learning general features of manipulated videos.
- FakeAVCeleb: A dataset focusing on audio-visual deepfakes, where a celebrity's face and voice are swapped. This dataset introduces complex scenarios of lip-sync and voice inconsistency, ideal for training the multimodal aspects of the system.
- KoDF (Korean Deepfake Dataset): A deepfake video dataset with a focus on Korean speakers, included to ensure linguistic and demographic diversity in training data. Incorporating KoDF helps the model generalise across different languages and ethnic groups.

These datasets collectively provide a wide range of conditions for the model to learn from, improving its resilience to new and unseen deepfakes. In addition to the above, a user-contributed dataset (with videos and audio from over 350 participants) may be integrated later to inject real-world variability and authenticity. This community-sourced data will only be used upon obtaining formal ethical approval from the university's review board, given the sensitive nature of personal media.

Ethical considerations are central to the project's design and execution. All datasets employed are either publicly available with proper licenses or have been cleared by an ethics committee. For any new data collected (such as the user-contributed set), participants provide informed consent, and all media is anonymised and securely stored. The system strictly operates as a detection and verification tool; notably, it does not generate any synthetic media and has no functionality to create deepfakes. This deliberate design choice aligns with responsible AI practices and avoids contributing to the very problem it aims to solve. Moreover, the development process will include routine audits for fairness and bias. The model's performance will be monitored across different demographic groups to ensure it does not disproportionately fail on certain ethnicities, accents, or genders. By complying with the General Data Protection Regulation (GDPR) and institutional ethical guidelines, the project prioritises the privacy and rights of individuals appearing in datasets. Overall, the approach balances innovation in deepfake detection with a commitment to ethical integrity.

Evaluation Strategy

The framework's effectiveness will be evaluated through a combination of quantitative metrics and qualitative analyses. Quantitative evaluation will involve standard classification metrics: accuracy, precision, recall, and F1-score, computed on a held-out test set from each core dataset. Given the importance of real-world conditions, the system's robustness to common video distortions (compression artefacts, resolution loss, noise) will also be tested. Each of the three primary datasets (DFDC, FakeAVCeleb, KoDF) will serve as both training and testing grounds in different experiments to assess cross-dataset generalisation –

for instance, training on two datasets and testing on the third to evaluate how well the model adapts to novel data. High performance across diverse datasets and resilience to artefacts will indicate a strong generalizable detector.

Qualitative evaluation will focus on the system's usability and transparency. A small user study is planned, wherein participants (e.g., journalists or media forensics students) will use the tool on a set of videos and provide feedback. This study will examine whether the explainability features (heatmaps, attention highlights) make the detection results understandable, and how much trust users place in the system's judgments. Metrics here may include user-reported clarity of explanations, perceived trustworthiness of results, and the time taken for a user to decide with the tool's aid. By collecting this feedback, the project can validate that the framework is not only accurate on paper but also effective and trustworthy from an end-user perspective.

Throughout development, a baseline comparison will also be performed: for example, comparing the hybrid CNN-transformer model against a standalone CNN classifier on the same data. This helps quantify the gains from the chosen architecture. Finally, all evaluation findings (both quantitative scores and qualitative insights) will be documented to identify strengths and areas for improvement, guiding future extensions of the work.

Expected Impact

If successful, the Reel or Real? The framework is expected to advance the state-of-the-art in deepfake detection and have a meaningful practical impact. Academically, it will demonstrate the value of combining multimodal analysis with explainability and provenance checks in a single system. This could encourage further research into holistic detection approaches that treat videos not just as pixels, but as complex audiovisual and contextual signals. By addressing key gaps in generalisation and transparency, the project sets a precedent for building detection models that are both robust and interpretable.

In real-world applications, the framework could serve as a powerful tool for digital media verification. For example, news organisations and social media platforms might deploy such a system to automatically screen user-uploaded videos for potential deepfakes, flagging suspicious content before it spreads. Law enforcement and fact-checking agencies could use it to verify the authenticity of video evidence. The inclusion of provenance analysis means the system not only catches AI-manipulated fakes but can also certify authentic content, which is equally important for establishing trust.

Ultimately, the project aims to bolster public trust in digital media by making deepfake detection more reliable and transparent. The expected outcome is a deployable prototype that proves AI-driven video verification can be done in an ethically responsible manner. By integrating technical rigour with user-centric design, Reel or Real? could help inoculate society against malicious deepfake campaigns and contribute to safer, more trustworthy online information ecosystems.

References

Tariq et al. (2025) - Introduce DF-P2E, a deepfake detection system that not only classifies videos but also explains its reasoning. It combines a CNN-based visual detector with Grad-CAM saliency maps, a captioning module, and a refined narrative generator powered by a fine-tuned LLM. This creates an explanation pipeline that highlights manipulated regions and provides natural-language reasons for flagging a clip. Methodologically, it is tested on the new DF40 dataset using detection metrics and user studies, achieving competitive accuracy while enhancing user understanding and trust. Relevance-wise, this work shows how explainability in deepfake detection can bridge the gap between black-box models and end-user trust, aligning with our focus on explainable AI and user-centric design, especially through Grad-CAM and narrative explanations.

Momin et al. (2025) - Explainable Deepfake Detection Across Modalities: This recent review offers a detailed overview of innovative deepfake detection techniques with an emphasis on their explainability across image, video, audio, and text domains. Unlike earlier surveys, it focuses specifically on how models provide explanations for their decisions, discussing methods such as Grad-CAM, LIME, SHAP, LRP, saliency maps, and Anchors in the context of multimodal deepfake detection. Main Contribution: It highlights the strengths and weaknesses of current methods and suggests future research directions to improve interpretability. For instance, while visual heatmaps are frequently used for image/video deepfakes, the authors point out that these are often not enough for non-experts without extra context. Relevance: The paper underlines the importance of explainability for our project, justifying our use of explanation modules and suggesting which techniques- like Grad-CAM for vision or SHAP/LIME for audio could be incorporated. It also emphasises the need to balance detection accuracy with interpretability, which is a key aspect of our context-aware framework.

Alsaeedi et al. (2025) - Audio-Visual Deepfake Detection via Emotional Consistency: This study proposes an audio-visual multimodal fusion approach that leverages emotion recognition to catch deepfakes. Methodology: The authors fuse pre-trained features from audio, visual, and textual modalities, focusing on the consistency of expressed emotions: audio tone, facial expressions, and even the sentiment of spoken words. The hypothesis is that fake videos often exhibit subtle mismatches in emotional tone (e.g., a smiling face but an angry voice). Their framework measures whether the sentiment of speech aligns with facial and vocal emotions, enhancing detection robustness. Using the FakeAVCeleb audio-visual dataset, the model achieved 95.24% accuracy, outperforming fusion models that ignore such sentiment alignment. Relevance: This work highlights the importance of contextual consistency (here, emotional context) in deepfake detection. It supports our project's context-aware multimodal strategy: by analysing audio and visual streams together (and potentially their semantic content), we can detect anomalies that single-modal models miss. It also informs our use of audio features (MFCC and beyond), suggesting that semantic or emotional cues can provide powerful signals for verification.

Zhang et al. (2025) - ERF-BA-TFD+: SOTA Multimodal Deepfake Detector: Zhang and colleagues present ERF-BA-TFD+, a cutting-edge deepfake detection model that won the 2025 deepfake detection challenge (DDL-AV audio-visual track) by a considerable margin. Main Innovation: The model introduces an Enhanced

Receptive Field (ERF) module to a multimodal architecture, enabling it to model long-range temporal dependencies across audio and video streams. In practice, ERF-BA-TFD+ processes video frames and corresponding audio spectrograms simultaneously, using a transformer-based backbone to capture subtle cross-modal inconsistencies (e.g., slight lip-sync delays or mismatched phonemes) that span longer durations. Results: On the challenging DDL-AV benchmark (which includes full-length videos with misaligned audio), it achieved state-of-the-art accuracy and efficiency, outperforming prior methods on metrics like precision and F1-score. This system's ability to handle asynchronous forgeries (e.g., fake audio dubbing over real video, and vice versa) demonstrates robustness to complex deepfake scenarios. Relevance: ERF-BA-TFD+ validates the efficacy of a hybrid CNN/Transformer design for multimodal deepfake detection – a similar approach to our proposed architecture. Its success underscores the importance of modelling temporal context and audio-visual synchronisation, reinforcing our decision to integrate an audio pipeline and sequence modelling. It also exemplifies how generalisation can be improved by training on diverse forgeries and using architectures that capture long-term dependencies, guiding our implementation to consider longer video contexts.

Romero-Moreno (2025) - Legal Frameworks, Ethics, and Provenance in Deepfake Detection: This interdisciplinary work examines deepfake detection through both a technical and a legal/ethical lens. It reviews key mitigation strategies – including AI-based detection, provenance tracking, and digital watermarking – emphasising the emerging C2PA standard for media authentication metadata. It also explores how deepfakes intersect with issues of evidence in court, privacy, free expression, and bias. Findings: The paper argues that purely technical solutions are insufficient unless paired with a supportive legal framework. It advocates for international coordination (citing UN Resolution 78/265) and interoperable standards to ensure deepfake detectors and provenance tools are widely adopted. Notably, it calls for regulations that encourage Explainable AI (XAI) and the adoption of C2PA metadata in content creation as part of a trustworthy AI ecosystem. Relevance: Romero-Moreno's analysis reinforces the ethical AI aspects of our project. It validates our inclusion of cryptographic provenance watermarking and C2PA metadata checks as forward-looking features for verification. Moreover, it highlights the importance of transparency (explainability) not just for users but for broader accountability and legal defensibility of our system's decisions. Incorporating these principles ensures our framework is not only technically sound but also aligned with emerging standards and ethical best practices in the fight against deepfakes.